INSTALLING WINDOWS SERVER 2016 LAB REPORT

CIS 227 — Server Administration & Network Security

BACKGROUND	
Course:	CIS 227 - Server Administration & Network Security
Term:	Spring 2026
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Lab Title:	Installing Windows Server 2016 Standard and Datacenter (Desktop Experience)
Week / Chapter Reference:	Week 1 / Chapter 1 - Installing Windows Server 2016
Assignment Reference:	Windows Server 2016 installation, first-login validation, and administrative analysis
Lab Type:	Windows Server Installation Lab Report
Date Completed:	2026-04-17

LAB OVERVIEW

Lab Scope:

- This report documents two Windows Server 2016 Desktop Experience installations completed in Oracle VM VirtualBox: Windows Server 2016 Standard and Windows Server 2016 Datacenter.
- The work included VM creation, storage selection, RAM and CPU assignment, virtual disk creation, ISO attachment, boot from media, operating system deployment, Administrator configuration, first login, and initial Server Manager validation.
- The finished systems were then used for comparative discussion covering edition differences, installation models, upgrade awareness, network-infrastructure planning, Features On Demand, and the security and resource implications of Server Core.

Platforms and Systems:

- The host system used for the lab was a Windows workstation running Oracle VM VirtualBox.
- The guest operating system for both documented installations was Windows Server 2016 deployed from evaluation media.
- The virtual machine used during the documented work was named CIS227-WS2016.
- The machine was configured with 4096 MB of RAM, 2 virtual CPUs, a 64 GB VDI disk, and NAT networking.
- Both Desktop Experience builds reached a successful first-login state and opened Server Manager correctly.

Files, Media, and Supporting Material:

- The installation media used for the lab was Windows_Server_2016_Eval.ISO.
- The ISO was stored at D:\Spring_2026_Course_Vaults\CIS 227 - Server Admin\06 Resources \00_ISOs\Windows_Server_2016_Eval.ISO.
- The automation script used for repeatable VM creation was New-CIS227-WS2016-VM.ps1, stored under the course vault script folder.
- That script first searched for VBoxManage.exe in standard VirtualBox locations, then searched under Program Files, and stopped with a clear error if VirtualBox was not installed.
- It then checked the exact ISO path above, searched the broader course vault if needed, and stopped with an error if Windows_Server_2016_Eval.ISO could not be found.

Learning Objectives:

- Identify the major Windows Server 2016 installation options, including Server Core and Desktop Experience.
- Recall the supported upgrade paths from earlier Windows Server versions.
- Explain the security and resource-efficiency advantages associated with Server Core installations.
- Describe how Features On Demand conserves disk space by removing unused role and feature files from the operating system.
- Install Windows Server 2016 in the appropriate environment while recognizing the distinction between Server Core and Desktop Experience deployment options.
- Examine existing network infrastructure and hardware compatibility to help determine the most appropriate upgrade path.
- Compare Server Core and GUI-based installations in terms of security, performance, and administrative requirements.

Administrative Importance:

- A stable installation baseline is necessary before roles, features, policies, services, storage, and security controls can be administered effectively.
- The completed baseline included edition selection, installation model, disk layout, network configuration, Administrator setup, and first-login validation.
- The script-based setup enforced consistent VM settings, folder creation, duplicate-name checks, storage attachment, and a repeatable start process.

SECTION A — WINDOWS SERVER 2016 STANDARD (DESKTOP EXPERIENCE)

Preparation and Virtual Machine Build:

- The Standard installation began with review of the assigned workflow so the build sequence matched the expected Windows Server 2016 process.
- The PowerShell automation script validated VBoxManage.exe, verified the ISO path, created the VM storage folder when needed, and stopped if a VM named CIS227-WS2016 was already registered.
- The VM was then created and registered with 4096 MB RAM, 2 virtual CPUs, a 64 GB VDI disk, NAT, IO APIC enabled, Intel 82540EM NIC emulation, and boot order set to DVD then disk.
- The script created the VDI, added SATA and IDE controllers, attached the virtual hard disk, mounted the Windows Server 2016 ISO as a DVD drive, and started the VM.

Standard Desktop Experience Installation Workflow:

1) The virtual machine was started and booted from the Windows Server 2016 installation media.
2) The Windows Setup environment loaded successfully, allowing the installation sequence to begin.
3) The language, time, and keyboard stage was passed and the installation sequence moved forward into the operating system deployment process.
4) Because evaluation media was used, the installation sequence was continued without a permanent product key requirement.
5) Windows Server 2016 Standard (Desktop Experience) was selected as the target build for this path.
6) The license step was accepted, the custom installation path was used, and the primary virtual disk was selected for deployment.
7) The server then progressed through file copy, file preparation, feature installation, and update phases.
8) After those phases completed, the system reached the Customize settings screen for the built-in Administrator account.
9) The built-in Administrator account was configured with the password N**********-2026!Z****
10) After setup completed and the machine rebooted, the server entered the full Desktop Experience environment successfully.
11) A successful first login was completed, the preferred network discovery choice was applied, and Server Manager Dashboard opened, confirming an operational server state.

Configuration and Validation Results:

- The completed installation path documented in this section is Windows Server 2016 Standard (Desktop Experience).
- The VM booted from ISO correctly, advanced through the expected setup stages, reached built-in Administrator configuration, and then completed first login successfully.
- The presence of the full graphical shell and Server Manager Dashboard verifies that the completed build used Desktop Experience rather than Server Core.
- The successful first-login state confirmed that the Standard build was ready for continued administrative work.

Standard Desktop Experience Screenshots:

Figure A1 - Standard installation nearing completion.

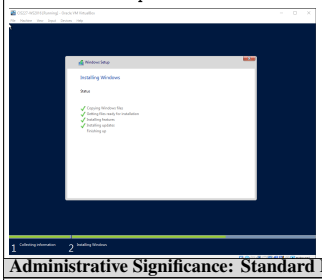


Figure A2 - Standard Administrator account setup.

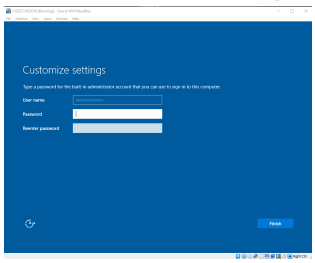
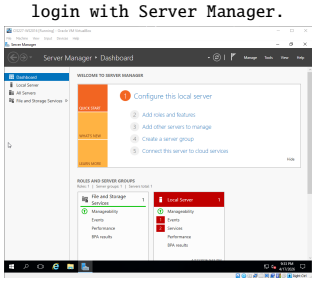


Figure A3 - Standard first login with Server Manager.



Administrative Significance: Standard Build

- The Standard Desktop Experience build established a stable and fully accessible administrative baseline for later course tasks.
- Because the machine reached first login and opened Server Manager successfully, the build was confirmed to be ready for continued configuration, role installation, and system management.
- The standardized script-driven VM build also made the Standard path easier to reproduce and document consistently.

SECTION B — WINDOWS SERVER 2016 DATACENTER (DESKTOP EXPERIENCE)

Preparation and Virtual Machine Build:

- The Datacenter installation followed the same structured VM-build approach so the results could be compared cleanly against the Standard path.
- The same automation logic was reused: VBoxManage validation, ISO verification, VM-folder creation, duplicate-name protection, and consistent VirtualBox hardware settings.
- The VM configuration remained aligned across both installations, including memory allocation, processor count, virtual disk size, storage location, NAT networking, and ISO-based boot.
- This consistency kept the main difference centered on edition choice rather than on large environmental changes.

Datacenter Desktop Experience Installation Workflow:

1) The Datacenter installation began by booting the virtual machine from the Windows Server 2016 ISO.
2) The Windows Setup environment loaded successfully and the installation sequence began normally.
3) The language, time, and keyboard stage was passed and the installation sequence moved into operating system deployment.
4) Because evaluation media was used, the installation workflow proceeded without a permanent product key requirement.
5) Windows Server 2016 Datacenter (Desktop Experience) was selected as the target build for this path.
6) The license step was accepted, the custom installation path was used, and the primary virtual disk was selected for deployment.
7) The system advanced through file copy, preparation, feature installation, and update phases without a fatal installation failure.
8) The server then reached the Customize settings screen for the built-in Administrator account.
9) The built-in Administrator account was configured with the password N**********-2026!Z****
10) After setup completed and the machine rebooted, the server entered the full Desktop Experience environment successfully.
11) A successful first login was completed, the preferred network discovery choice was applied, and Server Manager Dashboard opened, confirming an operational server state.

Configuration and Validation Results:

- The completed installation path documented in this section is Windows Server 2016 Datacenter (Desktop Experience).
- The VM booted from ISO correctly, advanced through the expected setup stages, reached built-in Administrator configuration, and then completed first login successfully.
- The presence of the full graphical shell and Server Manager Dashboard verifies that the completed build used Desktop Experience rather than Server Core.
- The successful first-login state confirmed that the Datacenter build was also ready for continued administrative work.

Datacenter Desktop Experience Screenshots:

Figure B1 - Datacenter installation nearing completion.

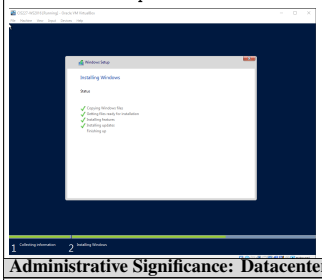


Figure B2 - Datacenter Administrator account setup.

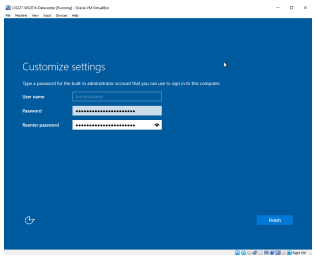
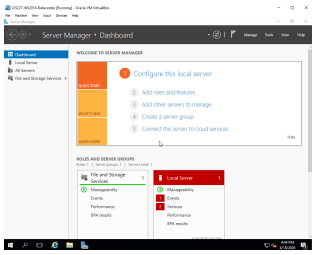


Figure B3 - Datacenter first login with Server Manager.



Administrative Significance: Datacenter Build

- The Datacenter Desktop Experience build established a second stable and fully accessible administrative baseline using the same operational workflow.
- Because the machine also reached first login and opened Server Manager successfully, the Datacenter path was validated as ready for continued configuration and management.
- The Datacenter path therefore supported a direct edition-level comparison without changing the fundamental VM build logic or validation method.

COMPARATIVE DISCUSSION, ADMINISTRATIVE ANALYSIS, LEARNING OUTCOMES, AND SUMMARY

Edition Comparison and Installation Observations:

- The Standard and Datacenter Desktop Experience builds followed the same broad installation sequence, including VM preparation, media boot, Windows Setup progression, Administrator configuration, and first-login validation.
- The major distinction between the two builds was the edition selected rather than the basic structure of the installation workflow itself.
- Because both builds used the same scripted VirtualBox baseline, the comparison stayed focused on the operating-system edition rather than on inconsistent hardware settings or storage choices.

Automation Script and VirtualBox Control Logic:

- New-CIS227-WS2016-VM.ps1 improved repeatability by validating prerequisites before any VM creation step was attempted.
- The script searched for VBoxManage.exe, checked the ISO location, enforced the planned VM specification, created required storage components, and attached both disk and DVD media.
- This fail-fast and repeatable setup reduced configuration drift and supported consistent testing across both Windows Server 2016 installation paths.

Installation Models, Resource Use, and Security Context:

- Desktop Experience provides a full graphical interface, which is especially useful during initial installation, navigation, and first-stage administrative learning.
- Server Core represents a more minimal installation model and matters in professional practice because it reduces GUI overhead, narrows the surface exposed to attack, and usually requires fewer system resources.
- From a security perspective, the reduced footprint of Server Core matters because fewer installed components usually means fewer services, fewer binaries, and fewer elements that must be patched and hardened.
- Features On Demand is also relevant because it allows unused role and feature files to be removed from disk so the operating-system footprint can be managed more efficiently.

Learning Outcomes:

- This lab reinforced that Server Core is a minimal installation model without the full graphical user interface, whereas Desktop Experience includes the complete graphical desktop and is generally easier to manage directly at the local console.
- Comparison of Server Core and Desktop Experience showed that Server Core is generally better for security and resource efficiency, while Desktop Experience is often better suited for easier hands-on administration and instructional use.
- Review of upgrade planning showed that the correct upgrade path depends on the Windows Server version already in use as well as the technical and administrative conditions of the environment.
- The lab further emphasized that existing network infrastructure, hardware compatibility, and broader organizational requirements must be examined before selecting an upgrade or deployment strategy.
- By completing the Windows Server 2016 installations in VirtualBox, practical experience was gained with the server deployment process and with evaluating how installation choices depend on the operating environment and administrative needs.

Upgrade Awareness, Infrastructure Planning, and Summary:

- Awareness of upgrade paths is important because existing Windows Server environments may require different planning decisions depending on the current version, infrastructure state, and goals.
- Examining network infrastructure also matters because the most appropriate deployment or upgrade path depends on compatibility, administrative requirements, security posture, and organizational context.
- The Standard Desktop Experience installation path was completed successfully, and the Datacenter Desktop Experience installation path was also completed successfully.
- In both cases, setup reached Administrator configuration, first login, and Server Manager access, so the resulting systems provided two usable Windows Server 2016 lab environments and a reliable technical record for later course work.